Supplementary Material for Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

S1. Empirical cross-term diagnostic

The variance correction of Eq. (8) assumes $\mathrm{Cov}(\tilde{g}_i^{\mathrm{c}}, g_m) \approx 0$, allowing the variance derivation to omit the cross-term. This assumption does not follow automatically from the fitting procedure because each JumpHMM uses the asset's full growth-rate history, including its market-correlated component. A remaining market loading in the generator draw would invalidate the closed-form scale. We tested the assumption by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real return path for a broad-market exchange-traded fund (ticker SPY):

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R} \sum_{r=1}^{R} \frac{\mathrm{Cov}((\tilde{g}_i^{(r)})^{\mathrm{c}},\, g_m)}{\sigma_{\mathrm{gen},i}\, \sigma_m}. \tag{S1}$$

Centering does not change covariance, so these values are numerically identical to those obtained from the raw draws. Per-ticker values were plotted as histograms in Supplementary Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composition methods), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composition method), and the Gaussian baseline had universe-mean normalized cross-covariances of 1.53×10^{-4}, 6.58×10^{-5}, and 2.26×10^{-4}, respectively. Their per-ticker values ranged from -0.0061 to 0.0061. These values quantify the cross-term omitted in Eq. (8).

S2. Bias correction for daily-rebalanced backtests

This appendix derives the bias correction in Eq. (S14). Under daily rebalancing, the raw synthetic ensemble had a median annualized growth rate about 22 percentage points above the long-run prior. The variance correction in Eq. (8) first sets each synthetic asset's second moment to its generator target. When that target matches the empirical second moment, the second-order formula for the diversification return is also matched at the asset level. The remaining difference can reflect omitted cross-asset residual covariance, higher-order Taylor terms, calibration drift, and the documented attenuation of realized diversification

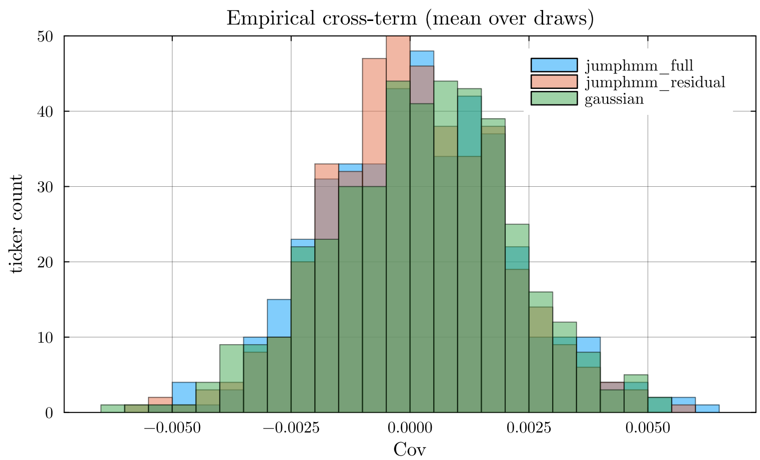

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}(\tilde{g}_i^c, g_m)/(\sigma_{\mathrm{gen},i}\,\sigma_m)$ **across the 423-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. Universe means were 1.53×10^{-4}, 6.58×10^{-5}, and 2.26×10^{-4} for the full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators.

returns (Bouchey et al., 2015). The location-shift correction absorbs these effects as a single empirical drag on log-wealth.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence. Each residual generator retains its own temporal structure, but the residual paths are generated independently across assets at each date. The construction therefore omits cross-asset residual covariance. Consider a daily-rebalanced portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t)\,\Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t \;=\; W_{t-1} \sum_i w_i \exp\bigl(g_i(t)\,\Delta t\bigr), \tag{S2}$$

so the per-step log return on the portfolio is:

$$\log\left(\frac{W_t}{W_{t-1}}\right) = \log\left[\sum_i w_i \exp\bigl(g_i(t)\,\Delta t\bigr)\right].$$

We suppress the time argument, writing g_i for $g_i(t)$. A second-order Taylor expansion of the exponential about zero gives:

$$\exp(g_i\,\Delta t) = 1 + g_i\,\Delta t + \tfrac{1}{2}(g_i\,\Delta t)^2 + O\bigl((g\,\Delta t)^3\bigr)\,.$$

Substituting this expansion into each summand gives:

$$\begin{aligned}\sum_i w_i \exp\bigl(g_i\,\Delta t\bigr) &= \sum_i w_i\bigl[1 + g_i\,\Delta t + \tfrac{1}{2}(g_i\,\Delta t)^2 \\ &\qquad\qquad + O\bigl((g\,\Delta t)^3\bigr)\bigr] \\ &= \sum_i w_i + \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 + O\bigl((g\,\Delta t)^3\bigr) \\ &= 1 + \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 + O\bigl((g\,\Delta t)^3\bigr). \qquad \text{(S3)}\end{aligned}$$

A second-order Taylor expansion of $\log(1+x)$ about $x = 0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1 + x$ with the right-hand side of Eq. (S3), the order-Δt correction is $x := \Delta t \sum_i w_i\,g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2(\sum_i w_i\,g_i)^2 + O((\Delta t)^3)$ because the cross term has an extra Δt factor. Substitution gives:

$$\begin{aligned}\log\frac{W_t}{W_{t-1}} &= \log\Bigl(1 + x + O\bigl((g\,\Delta t)^3\bigr)\Bigr) \\ &= x - \tfrac{1}{2}\,x^2 + O\bigl(x^3\bigr) \\ &= \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 \\ &\quad - \tfrac{1}{2}(\Delta t)^2\Bigl(\sum_i w_i\,g_i\Bigr)^2 + O\bigl((g\,\Delta t)^3\bigr) \\ &= \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2\Biggl[\sum_i w_i\,g_i^2 \\ &\qquad\qquad - \Bigl(\sum_i w_i\,g_i\Bigr)^2\Biggr] + O\bigl((g\,\Delta t)^3\bigr). \qquad \text{(S4)}\end{aligned}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1}\log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p \;=\; \sum_i w_i\,g_i \;+\; \tfrac{1}{2}\,\Delta t\Biggl(\sum_i w_i\,g_i^2 \;-\; \Bigl(\sum_i w_i\,g_i\Bigr)^2\Biggr) \;+\; O\bigl((\Delta t)^2\bigr). \qquad \text{(S5)}$$

Restoring the time argument, we now take unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i \;:=\; \mathbb{E}\bigl(g_i(t)\bigr), \qquad \sigma_i^2 \;:=\; \mathrm{Var}\bigl(g_i(t)\bigr), \qquad \text{(S6)}$$

the per-rate covariance $\Sigma_{ij} := \mathrm{Cov}\bigl(g_i(t), g_j(t)\bigr)$, and the per-rate portfolio variance $\sigma_p^2 := \mathrm{Var}\bigl(\sum_i w_i g_i(t)\bigr) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i\,g_i(t)^2$ and $(\sum_i w_i\,g_i(t))^2$. Both expand by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}\big(X^2\big) \;=\; \mathbb{E}\big(X\big)^2 \;+\; \mathrm{Var}(X). \tag{S7}$$

Applying this identity to $X = g_i(t)$ and summing against w_i gives:

$$\begin{aligned}
\mathbb{E}\Big(\sum_i w_i\, g_i(t)^2\Big) \;&=\; \sum_i w_i\, \mathbb{E}\big(g_i(t)^2\big) && \text{(linearity)}\\
&=\; \sum_i w_i \big[\, \mu_i^2 + \sigma_i^2 \,\big] && \text{(by Eq. (S7))}\\
&=\; \sum_i w_i\, \mu_i^2 \;+\; \sum_i w_i\, \sigma_i^2. && \text{(S8)}
\end{aligned}$$

Applying the identity to $X = \sum_i w_i g_i(t)$, whose mean is $\sum_i w_i \mu_i$ and whose variance is σ_p^2, gives:

$$\begin{aligned}
\mathbb{E}\Big(\big(\textstyle\sum_i w_i g_i(t)\big)^2\Big) \;&=\; \mathbb{E}\Big(\textstyle\sum_i w_i g_i(t)\Big)^2 \;+\; \mathrm{Var}\Big(\textstyle\sum_i w_i g_i(t)\Big) \quad \text{(by Eq. (S7))}\\
&=\; \Big(\textstyle\sum_i w_i\, \mu_i\Big)^2 \;+\; \sigma_p^2. && \text{(S9)}
\end{aligned}$$

We substituted Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouped the variance and mean contributions as follows:

$$\begin{aligned}
\mathbb{E}(g_p) \;&=\; \sum_i w_i\, \mu_i + \underbrace{\tfrac{1}{2}\,\Delta t\left(\sum_i w_i\, \sigma_i^2 - \sigma_p^2\right)}_{D}\\
&\quad + \tfrac{1}{2}\,\Delta t\left[\sum_i w_i\, \mu_i^2 - \Big(\sum_i w_i\, \mu_i\Big)^2\right] + O\big((\Delta t)^2\big),
\end{aligned} \tag{S10}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units (Booth and Fama, 1992; Markowitz, 1976). The middle term, D, is the diversification return: the $O(\Delta t)$ correction equal to half the difference between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2. The per-rate moments μ_i and σ_i^2 have units yr^{-1} and yr^{-2} respectively; the products $\Delta t\, \sigma_i^2$ and $\Delta t\, \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}). The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t\, \mu^2 \sim 3{\times}10^{-5}\ \mathrm{yr}^{-1}$ at $\mu \sim 0.08\ \mathrm{yr}^{-1}$ and negligible against the variance term $D \sim \Delta t\, \sigma^2 \sim 4{\times}10^{-2}\ \mathrm{yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T\,\Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2\sigma_i^2 + 2\sum_{i<j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i(1-w_i)\sigma_i^2 = \sum_{i<j} w_i w_j(\sigma_i^2+\sigma_j^2)$ that holds for any weights summing to one:

$$D \;=\; \tfrac{1}{2}\,\Delta t\left(\sum_i w_i\sigma_i^2 - \sigma_p^2\right) \;=\; \tfrac{1}{2}\,\Delta t\sum_{i<j} w_i w_j\ \mathrm{Var}(g_i-g_j)\;\geq\;0, \qquad \text{(S11)}$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i,j) comoves perfectly ($\mathrm{Var}(g_i-g_j)=0$) or all weight concentrates on a single asset. The rebalancing rule itself produces this effect: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz (1976) described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama (1992) called it the diversification return.

Substituting the single-index decomposition $g_i(t)=\alpha_i+\beta_i\,g_m(t)+\varepsilon_i(t)$, with market variance $\sigma_m^2 := \mathrm{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \mathrm{Var}(\varepsilon_i(t))$, and $\mathrm{Cov}(g_m,\varepsilon_i)=0$ (verified in Sec. S1), into Eq. (S11) gives:

$$\mathrm{Var}(g_i-g_j)\;=\;(\beta_i-\beta_j)^2\sigma_m^2\;+\;\mathrm{Var}(\varepsilon_i-\varepsilon_j), \qquad \text{(S12)}$$

Because g_m and the residuals are uncorrelated, this relation splits the diversification return into market-beta-dispersion and idiosyncratic components:

$$D \;=\; \underbrace{\tfrac{1}{2}\,\Delta t\,\sigma_m^2\sum_{i<j} w_i w_j(\beta_i-\beta_j)^2}_{D_{\mathrm{mkt}}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta t\sum_{i<j} w_i w_j\ \mathrm{Var}(\varepsilon_i-\varepsilon_j)}_{D_\varepsilon}, \qquad \text{(S13)}$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2}\,\Delta t\,\sigma_m^2[\sum_i w_i\beta_i^2 - (\sum_i w_i\beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t\,\sigma_m^2$. Across the 423-ticker universe the empirical β distribution spans $[0.03, 1.79]$ with interquartile range $[0.81, 1.18]$ and the SPY annualized variance $\Delta t\,\sigma_m^2 \approx 0.04\ \mathrm{yr}^{-1}$. The weighted variance of β is bounded by one quarter of its squared range, so D_{mkt} is at most $0.016\ \mathrm{yr}^{-1}$, or 1.6 percentage points per year over this universe. For the idiosyncratic component D_ε, residuals were approximately orthogonal across assets at the same time slice. Therefore, $\mathrm{Var}(\varepsilon_i-\varepsilon_j)\approx\sigma_{\varepsilon,i}^2+\sigma_{\varepsilon,j}^2$, and D_ε scales with the weighted-average annualized idiosyncratic variance $\Delta t\,\sigma_{\varepsilon,i}^2$ per name.

The second-order value of D depends on same-time unconditional moments, not on serial dependence. Equation (8) sets $\sigma_{\mathrm{gen},i}^2=\beta_i^2\sigma_m^2+\sigma_{\varepsilon,i}^2$ to its empirical target. Any residual generator with the same unconditional second moments therefore gives the same D. This includes a stationary first-order autoregressive [AR(1)] residual generator $\varepsilon_i(t)=\phi\,\varepsilon_i(t-1)+\eta_i(t)$ with independent and identically distributed innovations η_i tuned so that $\mathrm{Var}(\varepsilon_i)=\sigma_{\eta,i}^2/(1-\phi^2)=\sigma_{\varepsilon,i}^2$.

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample (IS): 2014–2024 ($T = 2{,}766$); out-of-sample (OoS): 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. We used the Jarque–Bera (JB) normality test and the Ljung–Box (LB) autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Standard deviation of G_t (% yr^{-1})	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	not used	not used
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	not used	not used
LB test on $\lvert G_t \rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	not used	not used

The parameter ϕ leaves the marginal distribution at a single time slice unchanged. It therefore does not change $\mathbb{E}(g_p)$ in Eqs. (S10) and (S13), but it changes sampling variance. For a stationary AR(1) residual with variance σ_ε^2, the variance of its sample mean is approximately $T^{-1}\sigma_\varepsilon^2(1+\phi)/(1-\phi)$. Under ergodicity, serial dependence changes per-path sampling noise, not the stationary mean. The raw median's $\sim$22-percentage-point excess over an external long-run growth prior is therefore not caused by i.i.d. sampling *across time* at the level of the second-order formula.

The direct structural difference is contemporaneous. The derivation assumes residuals are orthogonal across assets at each time slice, as they are in the independent per-asset generator. Real residuals share sector and style factors, which reduce $\mathrm{Var}(\varepsilon_i - \varepsilon_j)$ below the independent-generator value $\sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$. Matching each marginal variance does not restore this covariance. The synthetic D_ε will therefore exceed its real-market counterpart when the omitted covariance is positive. Measuring that contribution requires a joint-residual experiment, which we did not run. The gap may also include higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals have non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons (Bouchey et al., 2015). The location-shift correction of Eq. (S14) treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

To measure the bias, we applied a 20-asset, long-only target-weight allocator to the hybrid-composed universe. We calibrated the assets over 2014–2024, deployed the allocator in 2025, limited the universe to names with calibrated (α, β) and $\max \beta = 1.21$, and assumed a 5-basis-point round-trip cost. We then decomposed the path-mean annualized growth rate $\bar{g}_p \equiv (T\,\Delta t)^{-1} \log(W_{p,T}/W_{p,0})$ into static and rebalancing-driven components (Supplementary Table S2). The static-to-daily-rebalance switch increased the median by 27.73 percentage points. Adding a signal after daily rebalancing changed it by another 0.13 percentage points. The same allocator on the real 2025 growth-rate history produced 11.5%/yr,

Table S2: **Diversification-return decomposition on a 20-asset target-weight allocator deployed against the hybrid-composed 2025 universe.** Median path-mean annualized growth rate $\bar{g}_p$ across the synthetic ensemble. Equal-weight buy-and-hold serves as a passive baseline; the target-weight allocation is reported under static (no intra-year rebalance) and daily-rebalanced regimes, with and without a signal on top of the target weights. The static-to-daily increase of +27.73 percentage points on the same target weights isolates the diversification return of Eq. (S13). The same allocator deployed against the real 2025 growth-rate history is shown for comparison.

Allocation rule (synthetic 2025 universe)	$\bar{g}_p$ **(%/yr)**
Equal-weight buy-and-hold	7.68
Target-weight, static (no rebalance)	1.97
Target-weight, daily rebalanced	29.69
Target-weight, daily rebalanced + signal	29.83
Real 2025 history, same allocator	11.50

about 18 percentage points below the synthetic median. The synthetic residuals were independent across tickers and therefore omitted the contemporaneous covariance present in real markets. Determining how much of the gap this omission caused would require a joint-residual experiment with the marginals and allocator held fixed.

The correction subtracts a constant drift in log-wealth so that the ensemble median annualized growth rate matches an external prior $\bar{g}^{\text{prior}}$. At each time t, it multiplies every path by the same factor and therefore preserves cross-path wealth ranks. It also shifts every log-return increment by the same constant, which leaves the variance of those increments unchanged. For path p, we applied the following correction:

$$W_{p,t}^{\text{corr}} \;=\; W_{p,t} \cdot \exp\bigl(-\, b \,\cdot\, t\,\Delta t\bigr), \qquad b \;=\; \operatorname*{median}_{p} \bar{g}_p \;-\; \bar{g}^{\text{prior}}, \tag{S14}$$

with Δt the integration step in years and b, $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units. The correction is a linear detrend in log-wealth and remains multiplicative in wealth. It changes drawdown dates and depths, so it adjusts the return level rather than preserving drawdowns. It does preserve terminal-wealth ranks and the variance of log-return increments. We set $\bar{g}^{\text{prior}} = 8\%/\text{yr}$ before examining the holdout result. This value was based on external long-run equity-return estimates and an allowance for fees, turnover, and market impact (Damodaran, 2025; Ibbotson and Duff and Phelps, 2024). We used this documented prior rather than the realized test year to avoid circular calibration. Matching the realized year would force the synthetic median to equal the test sample. In this deployment, the real-2025 outcome of 11.5%/yr fell just above the median of the corrected ensemble.

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a tuning target and removes only a subset of the bias's consequences while leaving the underlying bias intact. Inflating per-trade transaction costs to absorb the

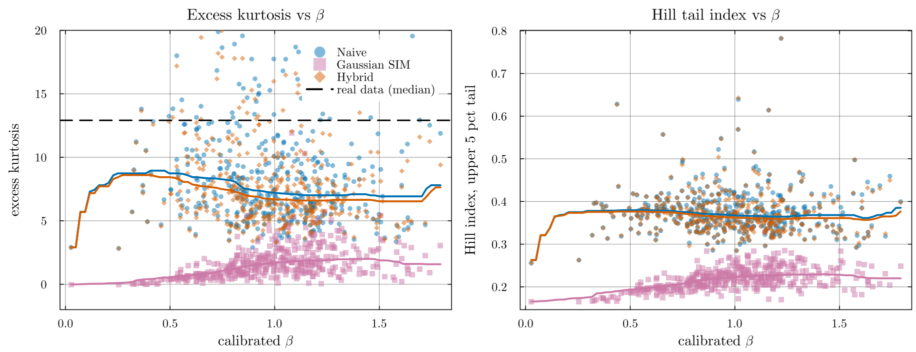

Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composition methods (orange and blue) track the generator's kurtosis, while the Gaussian single-index model (SIM) collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians remain near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22. The hybrid's median being below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (both retain the generator's tails) while Gaussian SIM does not.

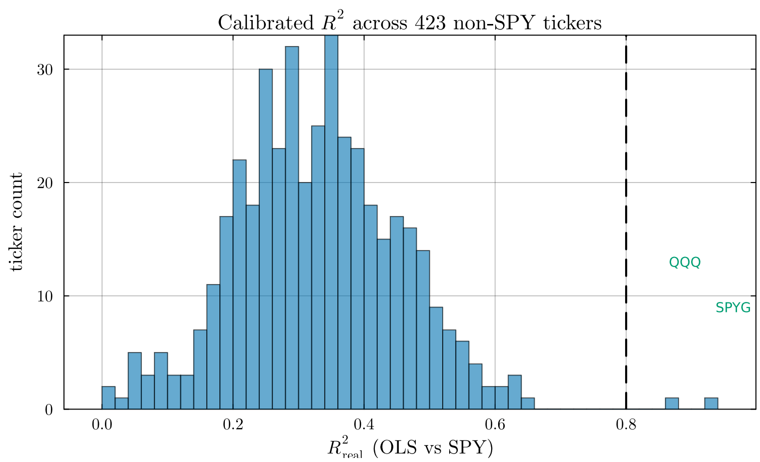

Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of ordinary least-squares (OLS) R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index exchange-traded funds (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) falls ~ 0.16 below the threshold. The R^2-preserving branch therefore contains only QQQ and SPYG in this universe.

diversification return would require round-trip costs of $\sim$70 basis points versus the 1–10 basis-point range assumed for liquid United States equity, shifting the misrepresentation from the synthetic generator to its cost model. Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, left the median Sharpe ratio essentially unchanged across the sweep, so standard-error-scale parameter drift did not absorb the bias. Imposing an AR(1) on the residual draw would conflict with the copula reorder step because reordering destroys temporal structure. Preserving both structures would require a block-ordered copula sampler and therefore a new residual model. The per-ticker block bootstrap (Politis and Romano, 1994) preserves each ticker's temporal residual structure but not covariance across tickers. A joint bootstrap or residual copula would be required for that purpose. The block bootstrap also replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the raw-versus-static deviation, $W^{\text{corr}} = W_{\text{static}} \cdot (W_{\text{raw}}/W_{\text{static}})^k$, hits the same target at k chosen to match the prior, but it changes the rank ordering of paths and inherits the static allocation's drawdown pattern.

The location-shift correction preserves percentile ranks and the relative ranking of strategies evaluated on the same paths. It also preserves centered moments of terminal log growth, including variance, skewness, and kurtosis. Because it changes the drawdown distribution, we evaluated drawdowns on the uncorrected ensemble. The correction does not repair residual autocorrelation, absolute Sharpe under daily rebalancing, or sensitivity to rebalancing frequency. Those targets require a residual model that preserves both temporal and cross-sectional dependence.

S3. Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov–Smirnov (KS) statistic between the empirical cumulative distribution functions (CDFs) of the observed and simulated growth-rate samples (G_{obs} of length n and G_{sim} of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ (Kolmogorov, 1933; Smirnov, 1948); the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson–Darling (AD) statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2 / [F(x)(1 - F(x))] \, dF(x)$, where F is the pooled empirical CDF (Anderson and Darling, 1952). The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \left| G_{\text{obs}}^{(i)} - G_{\text{sim}}^{(i)} \right|, \tag{S15}$$

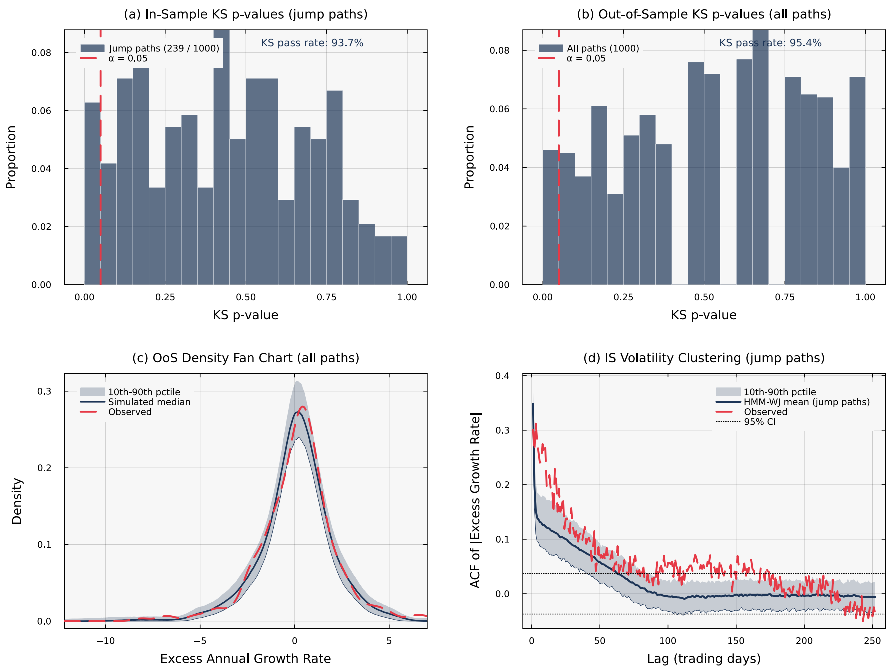

Figure S4: **Statistical validation of the hidden Markov model with jumps (HMM-WJ)** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p-values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p-values for all 1,000 paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) Observed and mean simulated in-sample autocorrelation function (ACF) of $|G_t|$ for jump paths.

with $G^{(i)}$ the ith order statistic. The Hill upper-tail index is the classical order-statistic estimator:

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log\left(\frac{X_{(i)}}{X_{(k)}}\right), \tag{S16}$$

with $X_{(1)} \geq X_{(2)} \geq \dots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α, $\hat{\xi} \approx 1/\alpha$. For the Hellinger calculation, we binned both samples on a common grid and normalized the bin counts to probability masses p and q. Their distance is:

$$H(p, q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_2 \in [0, 1], \tag{S17}$$

with $H = 0$ for identical probability masses and $H = 1$ for disjoint supports.

We abbreviate mean absolute error in the autocorrelation function as ACF-MAE.

S4. Derivation of the single-asset generator

This appendix derives the probability law of the fitted single-asset generator and connects each fitted component to the empirical feature it controls. The hidden Markov model with jumps (HMM-WJ), its no-jump restriction (HMM-NJ), and their symbols follow the main-text definitions in Eq. (2) and Algorithm 1. Unless noted otherwise, numerical values evaluate the closed-form expressions at the fitted SPY model of Table 1: $N = 100$ states, Student-t degrees of freedom $\nu = 5$, jump probability $\epsilon = 10^{-4}$, mean episode duration $\lambda = 90$ steps, tail-selection probability $p_{\text{neg}} = 0.52$, tail-set size $N_{\text{tail}} = 5$, and the in-sample series of $T = 2{,}766$ daily excess growth rates.

S4.1. Generative law

The no-jump model generates a hidden state path $S_0, \ldots, S_T$ on $\{1, \ldots, N\}$ and observations $G_1, \ldots, G_T$ with the joint density:

$$p(s_0, \ldots, s_T,\, g_1, \ldots, g_T) \;=\; \bar{\pi}_{s_0} \prod_{t=1}^{T} \mathbf{T}_{s_{t-1},\, s_t}\, f_{s_t}(g_t), \tag{S18}$$

where $\bar{\pi}_{s_0}$ is the initial probability of state s_0, $\mathbf{T}_{s_{t-1},s_t}$ is the probability of moving from state s_{t-1} to state s_t, and f_k is the emission density of state k. Under Eq. (2), f_k is the location-scale Student-t density:

$$f_k(g) \;=\; \frac{1}{\sigma_k}\,\psi_\nu\!\left(\frac{g-\mu_k}{\sigma_k}\right), \qquad \psi_\nu(x) \;=\; \frac{\Gamma\!\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\;\Gamma\!\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \tag{S19}$$

where ψ_ν is the density of a standard Student-t random variable with ν degrees of freedom, Γ is the gamma function, and (μ_k, σ_k) are the location and scale of state k. The factorization in Eq. (S18) states the structural property that drives everything that follows: given the state path, the observations are independent. Every form of serial dependence the generator can produce must therefore be carried by the state path (Rabiner and Juang, 1986).

The jump-augmented model enlarges the hidden state. Let $J_t \in \{0, 1, 2, \ldots\}$ count the forced steps remaining after time t, and let u be the forced-step state distribution, which mixes uniform draws over the two tail sets:

$$u_k \;=\; \begin{cases} p_{\text{neg}}/N_{\text{tail}} & k \in \mathcal{S}_-, \\ (1-p_{\text{neg}})/N_{\text{tail}} & k \in \mathcal{S}_+, \\ 0 & \text{otherwise.} \end{cases} \tag{S20}$$

Writing $p_K(m) = e^{-\lambda}\lambda^m/m!$ for the Poisson probability of episode length m and $\mathbf{1}\{\cdot\}$ for the indicator function, the pair (S_t, J_t) evolves by the transition kernel:

$$\begin{aligned} &\Pr\left(S_t = k,\, J_t = j \mid S_{t-1} = s,\, J_{t-1} = i\right) \\ &\quad= \begin{cases} u_k\,\mathbf{1}\{j = i-1\} & i \geq 1, \\ \left(1-\epsilon+\epsilon e^{-\lambda}\right)\mathbf{T}_{s,k}\,\mathbf{1}\{j=0\} \;+\; \epsilon\, p_K(j+1)\, u_k & i = 0, \end{cases} \end{aligned} \tag{S21}$$

and the emission draw depends only on the first coordinate, $G_t \sim f_{S_t}$. The two cases read as follows. While an episode is active ($i \geq 1$), the next state is an independent draw from u and the counter decrements; the tail set and the state within it are re-drawn at every forced step. While no episode is active ($i = 0$), the chain either takes an ordinary transition (no trigger, or a trigger whose Poisson draw returns zero) or starts an episode of length $j + 1 \geq 1$ whose first forced step occurs immediately. The pair (S_t, J_t) is therefore itself a Markov chain, and the HMM-WJ generator is a hidden Markov model on the enlarged state space $\{1, \ldots, N\} \times \{0, 1, 2, \ldots\}$. Algorithm 1 implements Eq. (S21) with the episode counter truncated at the simulation horizon; it emits the stationary initial draw as the first observation and applies the kernel from the second step onward, which changes the law of a path only through the absent possibility of a trigger at the first step. When an episode ends, ordinary transitions resume from the last forced tail state, so the chain re-enters the body of the state space through the tail rows of $\mathbf{T}$.

S4.2. Estimators and the one-step empirical law

The fitting procedure consists of four closed-form estimators, and each one matches a specific empirical object exactly. The Laplace partition comes first. For observations $G_1, \ldots, G_T$, the Laplace log-likelihood with location μ_L and scale b_L is:

$$\ell(\mu_L, b_L) \;=\; -T \ln(2b_L) \;-\; \frac{1}{b_L} \sum_{t=1}^{T} |G_t - \mu_L|. \tag{S22}$$

For any fixed $b_L > 0$, maximizing ℓ over μ_L minimizes $\sum_t |G_t - \mu_L|$, so $\hat{\mu}_L$ is the sample median. Setting the b_L derivative to zero at $\hat{\mu}_L$ gives:

$$\begin{aligned} \frac{\partial \ell}{\partial b_L} \;&=\; -\frac{T}{b_L} \;+\; \frac{1}{b_L^2} \sum_{t=1}^{T} |G_t - \hat{\mu}_L| \;=\; 0 && \text{(first-order condition)} \\ \hat{b}_L \;&=\; \frac{1}{T} \sum_{t=1}^{T} |G_t - \hat{\mu}_L|, && \text{(solve for } b_L\text{)} \end{aligned} \tag{S23}$$

the mean absolute deviation about the median. The state boundaries then invert the fitted Laplace cumulative distribution function (CDF) (Kotz et al., 2001). The Laplace CDF is piecewise exponential:

$$F_L(x) \;=\; \begin{cases} \frac{1}{2} \exp\big((x - \hat{\mu}_L)/\hat{b}_L\big) & x \leq \hat{\mu}_L, \\ 1 - \frac{1}{2} \exp\big(-(x - \hat{\mu}_L)/\hat{b}_L\big) & x > \hat{\mu}_L, \end{cases} \tag{S24}$$

and setting $F_L(Q_k) = k/N$ and solving each branch for Q_k gives the closed-form boundaries:

$$\begin{aligned} \frac{k}{N} \;&=\; \tfrac{1}{2}\, e^{(Q_k - \hat{\mu}_L)/\hat{b}_L} \implies Q_k \;=\; \hat{\mu}_L + \hat{b}_L \ln \frac{2k}{N}, && \text{(lower branch)} \\ \frac{k}{N} \;&=\; 1 - \tfrac{1}{2}\, e^{-(Q_k - \hat{\mu}_L)/\hat{b}_L} \implies Q_k \;=\; \hat{\mu}_L - \hat{b}_L \ln\Big(2 - \tfrac{2k}{N}\Big). && \text{(upper branch)} \end{aligned} \tag{S25}$$

The bins are equiprobable under the fitted Laplace, not under the empirical distribution, so the in-sample state counts are not uniform: across the 100 states, they range from 16 to 47 of the 2,766 observations. Every within-state mean μ_k and standard deviation σ_k was therefore computed from at least 16 observations, and no state required a fallback to the global moments.

The transition matrix is the maximum likelihood estimator (MLE) of a first-order Markov chain. Conditional on the assigned state sequence $s_1, \ldots, s_T$, let n_{jk} count the transitions from state j to state k and let $n_{j\cdot} = \sum_k n_{jk}$ be the row total. The likelihood of the transition parameters factorizes over rows into multinomials, and maximizing the log-likelihood subject to each row summing to one, with multiplier c_j for row j, uses the Lagrangian:

$$\mathcal{L} \;=\; \sum_{j,k} n_{jk} \ln \mathbf{T}_{jk} \;+\; \sum_j c_j \Big(1 - \sum_k \mathbf{T}_{jk}\Big), \tag{S26}$$

whose stationarity conditions solve in three steps:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{T}_{jk}} \;&=\; \frac{n_{jk}}{\mathbf{T}_{jk}} - c_j \;=\; 0 && \text{(first-order condition)}\\
\mathbf{T}_{jk} \;&=\; \frac{n_{jk}}{c_j} && \text{(solve for } \mathbf{T}_{jk}\text{)}\\
c_j \;&=\; \sum_k n_{jk} \;=\; n_{j\cdot} && \text{(row constraint } \textstyle\sum_k \mathbf{T}_{jk} = 1\text{)}\\
\widehat{\mathbf{T}}_{jk} \;&=\; \frac{n_{jk}}{n_{j\cdot}}. && \text{(counted estimator)} \qquad \text{(S27)}
\end{aligned}$$

The interior condition applies to cells with positive counts; for a cell with $n_{jk} = 0$ the log-likelihood is non-increasing in $\mathbf{T}_{jk}$, so its maximizer is the boundary value $\widehat{\mathbf{T}}_{jk} = 0$, which the same formula returns. Multiplying $\widehat{\mathbf{T}}_{jk}$ by the row frequency $n_{j\cdot}/(T-1)$ recovers $n_{jk}/(T-1)$, the empirical frequency of the state pair (j,k). The counted estimator therefore reproduces the in-sample joint law of (S_{t-1}, S_t) exactly; everything it asserts about longer horizons is the Chapman–Kolmogorov extension $\mathbf{T}^{\tau}$ of that one-step law (Bilmes, 1998; Grinstead and Snell, 1997). The emission estimator completes the pair: within each state, μ_k and σ_k are the sample mean and standard deviation of the assigned observations, and σ_k enters Eq. (S19) as the scale, so the emission standard deviation is $\sigma_k\sqrt{\nu/(\nu-2)}$. The aggregate consequence of this scale convention is quantified in Eq. (S31).

The initial distribution ties the pieces together. Let n_k count the occupancy of state k over the T observations, and let $\hat{\pi}_k = n_k/T$ be the occupancy frequency. The row total satisfies $n_{j\cdot} = n_j - \mathbf{1}\{s_T = j\}$, and the column total satisfies $n_{\cdot k} = n_k - \mathbf{1}\{s_1 = k\}$, because only the last observation opens no transition and only the first closes none. Substituting both identities gives:

$$
\begin{aligned}
\left(\hat{\pi}\,\widehat{\mathbf{T}}\right)_k &= \sum_j \frac{n_j}{T}\cdot\frac{n_{jk}}{n_{j\cdot}} && \text{(definitions)}\\
&= \sum_j \frac{n_{j\cdot}}{T}\cdot\frac{n_{jk}}{n_{j\cdot}} + \sum_j \frac{\mathbf{1}\{s_T=j\}}{T}\cdot\frac{n_{jk}}{n_{j\cdot}} && (n_j = n_{j\cdot} + \mathbf{1}\{s_T=j\})\\
&= \frac{n_{\cdot k}}{T} + \frac{\widehat{\mathbf{T}}_{s_T,k}}{T} && \text{(sum transitions into } k\text{)}\\
&= \hat{\pi}_k + \frac{\widehat{\mathbf{T}}_{s_T,k} - \mathbf{1}\{s_1=k\}}{T}. && (n_{\cdot k} = n_k - \mathbf{1}\{s_1=k\})
\end{aligned}
\tag{S28}
$$

The occupancy frequencies therefore satisfy the stationarity equation $\hat{\pi}\widehat{\mathbf{T}} = \hat{\pi}$ up to a fixed-point residual bounded by $1/T$. For the fitted SPY model, the solved eigenvector $\bar{\pi}$ deviates from the occupancy frequencies by at most 3.7×10^{-4}, the same order as $1/T$, against occupancy values between 0.0058 and 0.017. The initial distribution is consequently not an abstract spectral object: $\bar{\pi}$ is, up to a correction of the order of $1/T$, the vector of in-sample state frequencies.

S4.3. Stationary distribution and moments

The stationary observation law of the no-jump generator follows directly from Eq. (S18). Marginalizing the state at a single time gives the mixture density and CDF:

$$
p(g) = \sum_{k=1}^{N} \bar{\pi}_k\, f_k(g), \qquad F_{\text{mix}}(x) = \sum_{k=1}^{N} \bar{\pi}_k\, \Psi_\nu\!\left(\frac{x-\mu_k}{\sigma_k}\right), \tag{S29}
$$

where Ψ_ν is the CDF of the standard Student-t with ν degrees of freedom. Every ingredient of Eq. (S29) is an empirical summary: the weights are the in-sample state frequencies, and the component locations and scales are the within-state sample moments. The generator's marginal is therefore a 100-component semi-parametric estimate of the in-sample distribution, and reproducing the empirical marginal requires no further tuning. The mean is matched by construction. Weighting the within-state means by the occupancy frequencies telescopes into the grand mean:

$$
\begin{aligned}
\sum_k \hat{\pi}_k\,\mu_k &= \sum_k \frac{n_k}{T}\cdot\frac{1}{n_k}\sum_{t\,:\,s_t=k} G_t && \text{(definitions of } \hat{\pi}_k,\ \mu_k\text{)}\\
&= \frac{1}{T}\sum_{t=1}^{T} G_t, && \text{(each } t \text{ counted once)}
\end{aligned}
\tag{S30}
$$

and replacing $\hat{\pi}$ by $\bar{\pi}$ changes the value only through the $1/T$ correction of Eq. (S28): the model mean is 0.0636 yr^{-1} against the sample mean of 0.0631 yr^{-1}. The distributional match is equally direct. Evaluated at the fitted SPY model, the largest gap between the model CDF and the empirical CDF is $\sup_x |F_{\mathrm{mix}}(x) - \widehat{F}_T(x)| = 0.0045$, where $\widehat{F}_T$ is the in-sample empirical CDF. The two-sample Kolmogorov–Smirnov (KS) critical value at significance level $\alpha = 0.05$ for two samples of 2,766 observations is $1.358\sqrt{2/T} = 0.0365$, so the model-versus-data distance consumes about one eighth of the rejection threshold. Rejections of simulated paths are therefore driven by path-level sampling variation rather than by a misplaced marginal, which is consistent with the in-sample pass rates in Supplementary Table S4.

The second moment makes the role of the emission scale convention explicit. Conditioning on the state and applying the law of total variance gives:

$$\begin{aligned} \mathrm{Var}(G_t) &= \mathbb{E}\big[\mathrm{Var}(G_t \mid S_t)\big] + \mathrm{Var}\left(\mathbb{E}[G_t \mid S_t]\right) && \text{(law of total variance)} \\ &= \frac{\nu}{\nu-2}\sum_k \bar{\pi}_k\,\sigma_k^2 + \sum_k \bar{\pi}_k\,(\mu_k - \bar{\mu})^2, && \text{(Student-}t\text{ variance)} \end{aligned} \tag{S31}$$

where $\bar{\mu} = \sum_k \bar{\pi}_k \mu_k$ is the stationary mean. Define the within-state variance $V_{\mathrm{w}} = \sum_k \bar{\pi}_k \sigma_k^2$ and the between-state variance $V_{\mathrm{b}} = \sum_k \bar{\pi}_k(\mu_k - \bar{\mu})^2$. The corresponding empirical decomposition of the sample variance is $V_{\mathrm{b}} + V_{\mathrm{w}}$, up to degrees-of-freedom corrections, so the model variance exceeds the sample variance by $2V_{\mathrm{w}}/(\nu-2)$. At the SPY fit, $V_{\mathrm{w}} = 0.17\ \mathrm{yr}^{-2}$, only 3.6% of the total variance of $4.60\ \mathrm{yr}^{-2}$: the quantile partition concentrates almost all dispersion between states rather than within them. The inflation is correspondingly small, a model variance of 4.72 against $4.60\ \mathrm{yr}^{-2}$ (1.3% in standard deviation, 2.17 against $2.14\ \mathrm{yr}^{-1}$), and vanishes under Gaussian emissions ($4.61\ \mathrm{yr}^{-2}$).

The fourth moment explains the emission comparison in Supplementary Table S4. Write $G_t = \mu_k + \sigma_k X$ within state k, where X is a standard Student-t draw, and let $\delta_k = \mu_k - \bar{\mu}$. Expanding the fourth central moment and dropping the odd-power terms, which vanish by the symmetry of ψ_ν, gives:

$$\begin{aligned} \mathbb{E}\big[(G_t - \bar{\mu})^4\big] &= \sum_k \bar{\pi}_k\,\mathbb{E}\big[(\delta_k + \sigma_k X)^4\big] && \text{(condition on the state)} \\ &= \sum_k \bar{\pi}_k\left[\delta_k^4 + 6\,\delta_k^2\,\sigma_k^2\,\mathbb{E}(X^2)\right. \\ &\qquad\qquad \left. + \sigma_k^4\,\mathbb{E}(X^4)\right], \end{aligned} \tag{S32}$$

with the standard Student-t moments, finite for $\nu > 4$:

$$\mathbb{E}(X^2) = \frac{\nu}{\nu-2}, \qquad \mathbb{E}(X^4) = \frac{3\nu^2}{(\nu-2)(\nu-4)}. \tag{S33}$$

At $\nu = 5$ these values are 5/3 and 25: the within-state kurtosis is 9, against 3 for Gaussian emissions. The excess kurtosis of the stationary mixture is $\mathbb{E}[(G_t - \bar{\mu})^4]/\mathrm{Var}(G_t)^2 - 3$; evaluating it at the SPY fit gives 8.34 with Student-t emissions and 5.82 with Gaussian emissions, against an observed value of 7.72. The emission choice enters Eq. (S32) through both moments in Eq. (S33); the larger change by far is $\mathbb{E}(X^4)$ rising from 3 to 25, which acts on the σ_k^4 terms that the wide tail states dominate. The per-path simulated values reported in Supplementary Table S4 (7.5 and 5.5) sit below these population values because the sample kurtosis of a heavy-tailed series of 2,766 observations is biased downward, but the population gap of 2.5 between the emission choices matches the simulated gap of 2.0 in both direction and size.

S4.4. Dependence structure without jumps

The results stated so far concern single time slices; the dependence structure is where the no-jump generator falls short of the data, and the failure is structural rather than a matter of estimation. Within state k, the sojourn time is geometric: the chain remains for m steps with probability $\mathbf{T}_{kk}^{m-1}(1-\mathbf{T}_{kk})$ and therefore stays for $1/(1-\mathbf{T}_{kk})$ steps on average, between one and two days for every fitted state (Supplementary Fig. S5c). The autocorrelation of absolute returns inherits this short memory. Let $a_k = \mathbb{E}\big[\,|G_t| \,\big|\, S_t = k\big]$ be the within-state mean absolute return and $\bar{a} = \sum_k \bar{\pi}_k a_k$ its stationary average. Because observations are conditionally independent given the states, the lag-τ product moment reduces to a state-pair sum:

$$\begin{aligned}
\mathbb{E}\big[\,|G_t|\,|G_{t+\tau}|\,\big] \;&=\; \sum_{j,k} \Pr(S_t = j,\, S_{t+\tau} = k)\; a_j\, a_k \qquad &&\text{(conditional independence)}\\
&=\; \sum_{j,k} \bar{\pi}_j \left(\mathbf{T}^{\tau}\right)_{jk} a_j\, a_k, &&\text{(Chapman–Kolmogorov)}
\end{aligned} \tag{S34}$$

so the autocorrelation function (ACF) of $|G_t|$ is:

$$\mathrm{ACF}_{|G|}(\tau) \;=\; \frac{\sum_{j,k} \bar{\pi}_j \left(\mathbf{T}^{\tau}\right)_{jk} a_j\, a_k \;-\; \bar{a}^2}{\mathrm{Var}(|G_t|)}. \tag{S35}$$

As τ grows, $\mathbf{T}^{\tau}$ converges to its rank-one limit, every row equal to $\bar{\pi}$, at the geometric rate $|\theta_2|^{\tau}$, where θ_2 is the second-largest eigenvalue of $\mathbf{T}$ in modulus; the ACF is therefore bounded by a constant multiple of $|\theta_2|^{\tau}$. For the fitted SPY matrix, $|\theta_2| = 0.253$, and $|\theta_2|^{\tau}$ falls below 0.01 from lag 4 onward. Evaluating Eq. (S34) gives a model ACF of 0.26 at lag 1, close to the observed 0.30 because the counted matrix pins the one-step state-pair law, but 0.021 at lag 3 and effectively zero from lag 5, while the observed ACF remains near 0.26 at lag 5 and 0.14 at lag 25. No first-order chain fitted by transition counting can do otherwise: matching the one-step law fixes $\mathbf{T}$, and the spectral gap of that matrix then forces geometric mixing within a week. This is the quantitative content of the main-text statement that the counted transition probabilities leave extreme-return states too quickly to reproduce volatility clustering, and it is the failure the jump mechanism corrects.

S4.5. Jump-episode arithmetic

The jump mechanism overlays the fast-mixing chain with rare, long episodes, and its operating characteristics follow from renewal arguments. On each step with no active episode, an episode with at least one forced step begins with probability $p_e = \epsilon(1 - e^{-\lambda})$, since a trigger whose Poisson draw returns zero is discarded; conditional on starting, the episode length is the zero-truncated Poisson draw with mean $\mathbb{E}[K \mid K \geq 1] = \lambda/(1 - e^{-\lambda})$. At $\lambda = 90$, the correction $e^{-\lambda}$ is below 10^{-39}, so $p_e = \epsilon$ and $\mathbb{E}[K \mid K \geq 1] = \lambda$ to machine precision. The long-run fraction of forced steps, denoted π_J, follows from the renewal-reward theorem. A renewal cycle consists of the free steps preceding an episode and the episode itself: the number of free trials up to and including the trigger is geometric with mean $1/p_e$, the triggering trial is itself the first forced step, and the episode contributes $\mathbb{E}[K \mid K \geq 1]$ forced steps in total:

$$\begin{aligned}
\pi_J &= \frac{\mathbb{E}[K \mid K \geq 1]}{\dfrac{1-p_e}{p_e} + \mathbb{E}[K \mid K \geq 1]} && \text{(forced steps over cycle length)} \\
&= \frac{p_e\, \mathbb{E}[K \mid K \geq 1]}{1 - p_e + p_e\, \mathbb{E}[K \mid K \geq 1]} && \text{(multiply through by } p_e\text{)} \\
&= \frac{\epsilon\lambda}{1 - \epsilon + \epsilon e^{-\lambda} + \epsilon\lambda} && (p_e\, \mathbb{E}[K \mid K \geq 1] = \epsilon\lambda) \\
&\approx \frac{\epsilon\lambda}{1 + \epsilon\lambda} \;=\; 0.0089. && \text{(drop } \epsilon e^{-\lambda}\text{, then } \epsilon\text{)} \qquad \text{(S36)}
\end{aligned}$$

Fewer than one step in a hundred is forced at the fitted values; the measured fraction across 400 simulated in-sample-length paths is 0.0088. The product $\epsilon\lambda$ is the quantity that sets this fraction, a point that matters for identification in Section S4.6.

At the path level, the same arithmetic determines how episodes are distributed across the ensemble. A simulated path contains no episode only if every one of its $T-1$ transition steps fails to trigger one, so the jump-path probability at $T = 2{,}766$ is $1 - (1 - p_e)^{T-1} = 24.2\%$; the simulated ensemble gives 24.8% (99 of 400 paths), matching the $\sim 24\%$ jump-path fraction conditioned on in Supplementary Fig. S4. Episodes begin at the long-run rate of one per expected renewal cycle, so a path carries about 0.274 episodes in expectation. The expected gap between episodes is $1/\epsilon = 10^4$ trading days, about 40 trading years, so the ensemble is a two-population mixture: roughly three quarters of the paths never trigger an episode and follow the HMM-NJ law exactly, while the remaining quarter carry a single episode (about one jump path in eight carries a second). Within a jump path, the forced fraction is $\lambda/T = 3.3\%$ for a single episode; the simulated mean is 3.6%, slightly higher through the multi-episode paths. Forced steps differ sharply from free ones. Under the forced-step distribution u of Eq. (S20), the conditional standard deviation is 5.38 yr^{-1} against the free-step 2.17 yr^{-1}, a variance ratio of 6.1, and the mean absolute return is $\bar{a}_J = \sum_k u_k a_k = 4.96$ yr^{-1} against $\bar{a} = 1.47$ yr^{-1}. The stationary state marginal of the augmented chain is the mixture $(1 - \pi_J)$ times the free-phase distribution plus π_J times u; the free phase relaxes from the post-episode tail state back to $\bar{\pi}$ at the rate $|\theta_2|$ within a few steps. Using $\bar{\pi}$ as the initial law therefore misallocates $O(\pi_J) \approx 1\%$ of probability mass, which quantifies the main-text remark that $\bar{\pi}$ is exact for the ordinary transition matrix but approximate for the jump-augmented process.

S4.6. Autocorrelation and kurtosis under jumps

The tuning objective of Eq. (3) scores the absolute-return ACF and the excess kurtosis, and both statistics admit closed-form approximations that show what (ϵ, λ) control. The ACF contribution rests on one more renewal quantity: the probability $h(\tau)$ that, given a step is forced, the step τ later belongs to the same episode. A uniformly chosen forced step lands in an episode with the length-biased law, an episode of length m covering m forced steps, and its position within that episode is uniform. Combining the two gives:

$$\begin{aligned} h(\tau) \;&=\; \sum_{m\geq 1} \frac{m \,\Pr(K = m \mid K \geq 1)}{\mathbb{E}[K \mid K \geq 1]} \cdot \frac{(m-\tau)_+}{m} \\ &=\; \frac{\mathbb{E}\big[(K-\tau)_+ \,\big|\, K \geq 1\big]}{\mathbb{E}[K \mid K \geq 1]} && \text{(sum over } m\text{)} \\ &\approx\; \Big(1 - \frac{\tau}{\lambda}\Big)_+, && (K \text{ concentrated near } \lambda) \end{aligned} \tag{S37}$$

where $(x)_+ = \max(x, 0)$ and the approximation holds because the Poisson length concentrates within a few multiples of $\sqrt{\lambda} \approx 9.5$ around λ. The same-episode probability therefore decays almost linearly from one to zero over the horizon λ. The ACF follows by conditioning on the forced-step indicator $Z_t = \mathbf{1}\{\text{step } t \text{ is forced}\}$. Given the indicator path, every forced step's state is an independent draw from u, so two forced observations are conditionally independent; the dependence linking a forced observation to a later free one runs only through the episode's final state and, like the free-phase covariance itself, decays at the $|\theta_2|^\tau$ rate of Section S4.4. For lags beyond that scale, only the conditional means covary. Write $A_t = |G_t|$ for the absolute return. The conditional mean is $\mathbb{E}[A_t \mid Z_t] = \bar{a} + Z_t\,(\bar{a}_J - \bar{a})$, and for rare episodes the joint forced probability is $\Pr(Z_t = 1, Z_{t+\tau} = 1) \approx \pi_J\, h(\tau)$, so:

$$\begin{aligned} \mathrm{Cov}\big(A_t, A_{t+\tau}\big) \;&\approx\; \mathrm{Cov}\big(\mathbb{E}[A_t \mid Z_t],\, \mathbb{E}[A_{t+\tau} \mid Z_{t+\tau}]\big) && \text{(independence given } Z\text{)} \\ &=\; (\bar{a}_J - \bar{a})^2\, \mathrm{Cov}\big(Z_t,\, Z_{t+\tau}\big) && \text{(conditional mean in } Z\text{)} \\ &\approx\; (\bar{a}_J - \bar{a})^2\, \pi_J\, h(\tau), && \text{(episodes rare)} \end{aligned} \tag{S38}$$

so the jump mechanism adds an ACF component of height $\pi_J(\bar{a}_J - \bar{a})^2 / \mathrm{Var}(A_t)$ that decays almost linearly to zero at lag λ, precisely the slow component the counted chain cannot produce. For a jump path carrying a single episode, the relevant forced fraction is λ/T rather than π_J, and evaluating the resulting expression at the fitted values, with $\mathrm{Var}(A_t) = 3.01$ under the same conditioning, gives a predicted plateau of 0.13. Supplementary Table S3 compares this prediction with the mean simulated jump-path ACF: the two differ by 0.013 at lag 5 and by less than 0.01 from lag 10 onward. At lags 1 to 3 the simulated

Table S3: **Episode-overlap prediction of the jump-path absolute-return autocorrelation function (ACF) against simulation.** The prediction evaluates Eq. (S38) at the fitted SPY values with the single-episode forced fraction λ/T in place of π_J and the same-episode probability $h(\tau)$ of Eq. (S37). The simulated column is the mean ACF of $|G_t|$ over the 99 jump-active paths among 400 simulated in-sample-length paths. The free-phase component of Eq. (S34) is excluded from the prediction and has decayed below 0.01 at the lags shown.

Lag τ	$h(\tau)$	**Predicted ACF**	**Simulated ACF**
5	0.944	0.124	0.137
10	0.889	0.117	0.123
15	0.833	0.110	0.115
20	0.778	0.102	0.105
25	0.722	0.095	0.099

values exceed the episode component (0.35 against 0.13 at lag 1) because the free-phase geometric component of Section S4.4 adds on top before it dies out.

The kurtosis contribution pulls in the opposite direction. Treating a jump path as a two-regime mixture with weights $(1 - \lambda/T,\ \lambda/T)$ and evaluating the fourth-moment expression of Eq. (S32) with the forced-step distribution u in place of $\bar{\pi}$ for the episode regime gives a population excess kurtosis of 7.20 for a single-episode jump path, against 8.34 with no jumps. Forced steps inflate the variance, by about 17% at the fitted values, faster than they inflate the fourth moment, because u draws uniformly within each tail set rather than extending the extreme tail. Adding jump content therefore lowers kurtosis toward and past the observed value while it raises the slow ACF component; the jump-share sweep reported in the main text traces the same tradeoff empirically, with the Hill tail index flat across the sweep because the tail exponent is set by ν, not by the jump content. The objective of Eq. (3) balances these two effects.

The same expressions show how the two jump parameters are identified, and where they are not. Within a single-episode jump path, the ACF component of Eq. (S38) has height proportional to λ/T and horizon λ, and the kurtosis shift depends on the same forced fraction: every statistic entering Eq. (3) for such a path is controlled by λ. The parameter ϵ moves the objective only through the incidence of jump paths, which the jump-active conditioning removes, and through the small fraction of multi-episode paths. This weak dependence is consistent with the tuned $\epsilon = 10^{-4}$ lying at the lower edge of the explored range for SPY and for all three cross-asset tickers, and with the Monte Carlo sensitivity of the tuned optima noted in Section S8. The objects entering every expression in this appendix, the fitted boundaries, transition matrix, and residence times, are plotted in Supplementary Fig. S5.

S5. Transition matrix internals and partition diagnostics

Supplementary Fig. S5 shows the fitted hidden Markov model with jumps (HMM-WJ) for SPY with $N = 100$ states. The fitted Laplace cumulative distribution function (CDF) and the empirical CDF are shown with the 99 state

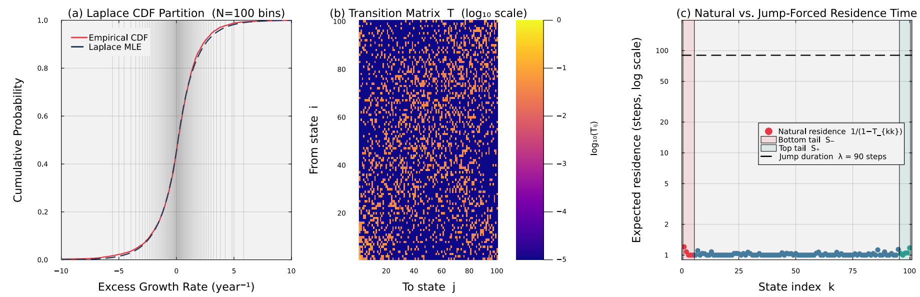

Figure S5: **Fitted hidden Markov model with jumps (HMM-WJ) internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace cumulative distribution function (CDF) on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the fitted mean jump duration $\lambda = 90$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

boundaries Q_k. The transition matrix $\mathbf{T}$ has a near-diagonal band, with most off-diagonal mass within ± 5 states of the diagonal. Under the ordinary Markov transitions, every state has an expected residence time of only 1–2 steps. The fitted jump duration is $\lambda = 90$ steps, about two orders of magnitude longer. Every state had at least one in-sample observation.

S6. Emission distribution and semi-Markov comparison

To compare the emission and duration specifications, we evaluated three variants on SPY (Supplementary Table S4): a hidden semi-Markov model (HSMM) baseline adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student-t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student-t emissions ($N = 100$). Both HMM-WJ variants had higher in-sample KS and AD pass rates than the HSMM-style baseline. The HMM-WJ variants use 100 states and the HSMM uses 8, so this comparison does not assign the pass-rate difference solely to the duration model. Holding the rest of HMM-WJ fixed, changing the emission from Normal to Student-t ($\nu = 5$) increased simulated excess kurtosis from 5.5 to 7.5, against an observed value of 7.7. The Student-t emission also increased the KS and AD pass rates, while ACF-MAE and the distributional distances changed modestly.

S7. State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for the hidden Markov

Table S4: **Three-way comparison of emission distributions and duration specifications for SPY.** Standard errors in parentheses; bold marks the best value per row. The HSMM is a hidden semi-Markov-style model adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student-t emissions); HMM-WJ: hidden Markov model with jumps ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$. Kolmogorov–Smirnov (KS), Anderson–Darling (AD), and mean absolute error in the autocorrelation function (ACF-MAE) are defined in Section S3.

Metric	**HSMM (Student-t)**	**HMM-WJ (Gaussian)**	**HMM-WJ (Student-t)**
States	$K = 8$	$N = 100$	$N = 100$
In-sample (2,766 trading days, 2014–2024)			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	**98.3** (0.4)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	**94.8** (0.7)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	**7.5** (0.09)
ACF-MAE	0.059 ($<$0.001)	**0.052** ($<$0.001)	0.053 ($<$0.001)
Wasserstein-1	0.176 (0.001)	0.101 (0.002)	**0.097** (0.001)
Hellinger	0.113 ($<$0.001)	0.076 ($<$0.001)	**0.074** ($<$0.001)
Out-of-sample (249 trading days, 2025)			
KS pass rate (%)	**96.2** (0.6)	95.0 (0.7)	95.4 (0.7)
AD pass rate (%)	**96.7** (0.6)	95.9 (0.6)	95.3 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	**6.2** (0.15)
ACF-MAE	0.042 ($<$0.001)	**0.040** ($<$0.001)	**0.040** ($<$0.001)
Wasserstein-1	0.287 (0.002)	**0.275** (0.005)	**0.275** (0.005)
Hellinger	0.239 (0.001)	0.212 (0.001)	**0.210** (0.001)

Table S5: **Hybrid composition method broken out by branch.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2-preserving branch (QQQ and SPYG) had median absolute errors of 0.005 for β and 0.001 for R^2, with a KS pass rate of 100%.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	73.5	0.250	7.142
`r2-preserve`	2	0.005	0.001	100.0	0.099	11.525

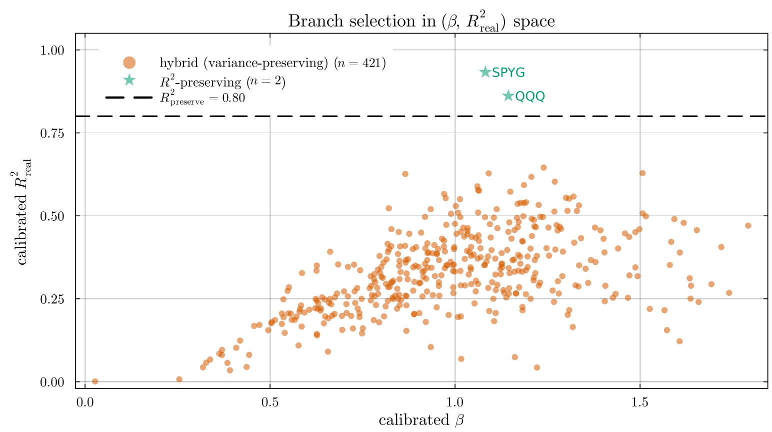

Figure S6: **Branch assignment in calibrated $(\beta, R^2_{\mathrm{real}})$ space.** The $R^2_{\mathrm{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2-preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\mathrm{gen},i}^2$ stays below 0.90 for every ticker.

model with no jumps (HMM-NJ) and HMM-WJ. Pass rates varied little across $N \in \{60, \ldots, 200\}$. At $N = 30$ both models lost several percentage points on the AD test. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined. We adopted $N = 100$ for all subsequent experiments.

S8. Cross-asset generalization

We repeated the fitting and evaluation procedure on three stocks: NVDA (semiconductors, high volatility), JNJ (health care, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at ($n_{\mathrm{paths}} = 200, \mathrm{seed} = 1234$) are reported in Supplementary Table S8. The objective combines ACF and kurtosis errors with fixed weights. Because kurtosis varies across finite samples of heavy-tailed paths, the selected grid point can change with the Monte Carlo sample. The reported (ϵ^*, λ^*) values are reproducible with the stated path count and seed, but they are not unique per-ticker optima. At the selected parameter values, in-sample HMM-WJ KS pass rates were 98.9% for NVDA, 99.4% for JNJ, and 99.1% for JPM. Out-of-sample KS pass rates on the 249-day 2025 hold-out were 97.7%, 74.0%, and 79.2%, respectively.

S9. SIM composition derivations and properties verification

The variance-corrected single-index composition states two closed-form results, the variance scale of Eq. (8) and the residual-variance target of Eq. (11). The

Table S6: **Aggregate metrics by β quartile, across composition methods.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines from 87.6% in Q1 to 65.9% in Q4 as the market loading ratio ρ_i grows; the naive KS pass rate falls from 24.6% to 1.3% over the same range.

β bucket	Method	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	24.6	0.403	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	87.6	0.177	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	1.9	0.621	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	74.9	0.231	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.7	0.792	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	66.1	0.279	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	1.3	0.965	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	65.9	0.339	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table S7: **Clipping-branch stress test under market-volatility scaling.** The market growth-rate path was rescaled by $\gamma \in \{1, 2, 3\}$ with the calibrated marginals held fixed, and the full hybrid composition method re-evaluated on the 423-ticker universe. As γ rises, the share of clipped tickers grows.

γ	**Clipped (%)**
1	0.0
2	76.4
3	95.2

Table S8: **Cross-asset generalization of the hidden Markov model with jumps (HMM-WJ) across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed = 1234, and the corresponding HMM-WJ Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates on 1,000 simulated paths at $\alpha = 0.05$ against each ticker's in-sample (IS; 2014–2024) and out-of-sample (OoS; 2025) growth-rate history. Values in parentheses are binomial standard errors (SEs) for pass rates. The (ϵ^*, λ^*) values are reported as the parameter values used for the IS / OoS metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	**IS KS (%)**	**IS AD (%)**	**OoS KS (%)**	**OoS AD (%)**
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (health care)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

derivation below gives the large-T OLS coefficients and composed variance. To derive Eq. (8), first center the full-return draw as in Eq. (6), then set $\varepsilon_i = s_i \tilde{g}_i^{\mathrm{c}}$ for $s_i \in (0, 1]$ and require that the composed series have the same variance as the generator path:

$$\mathrm{Var}(g_i) \;=\; \mathrm{Var}(\alpha_i + \beta_i\, g_m + \varepsilon_i) \;=\; \mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \sigma^2_{\text{gen},i}, \tag{S39}$$

where α_i drops out because it is constant. Expanding the sum gives:

$$\begin{aligned}\mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;&= \beta_i^2\; \mathrm{Var}(g_m) + 2\beta_i\; \mathrm{Cov}(g_m,\, \varepsilon_i)\\ &\quad + \mathrm{Var}(\varepsilon_i).\end{aligned} \tag{S40}$$

Substituting $\varepsilon_i = s_i\, \tilde{g}_i^{\mathrm{c}}$ gives the two moments:

$$\begin{aligned}\mathrm{Var}(\varepsilon_i) &= s_i^2\, \sigma^2_{\text{gen},i},\\ \mathrm{Cov}(g_m,\, \varepsilon_i) &= s_i\; \mathrm{Cov}(g_m,\, \tilde{g}_i^{\mathrm{c}}).\end{aligned}$$

Under the assumption $\mathrm{Cov}(\tilde{g}_i^{\mathrm{c}},\, g_m) \approx 0$, the cross term vanishes and the variance condition becomes $\beta_i^2\, \sigma_m^2 + s_i^2\, \sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$. Solving for s_i^2 gives Eq. (8).

To derive Eq. (11), write the population R^2 identity for Eq. (4):

$$R^2 \;=\; \frac{\beta_i^2\, \sigma_m^2}{\beta_i^2\, \sigma_m^2 + \sigma^2_{\varepsilon,i}}. \tag{S41}$$

Solving this identity for the residual variance gives Eq. (11). Because $\overline{\varepsilon}_i = 0$ after centering and scaling, the sample mean of a composed path obeys $\overline{g}_i = \alpha_i + \beta_i^{\text{eff}}\overline{g}_m$. Hence, the finite-sample OLS intercept was:

$$\hat{\alpha}_i = \overline{g}_i - \hat{\beta}_i \overline{g}_m = \alpha_i - (\hat{\beta}_i - \beta_i^{\text{eff}})\overline{g}_m. \tag{S42}$$

Under the zero-cross-covariance assumption, in the large-T limit OLS therefore gives $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$. Here $\beta_i^{\text{eff}} = \beta_i$ unless Eq. (10) clips the market loading. On the variance-preserving branch, the composed variance is $\sigma^2_{\text{gen},i}$. Without clipping, $\mathrm{Var}(g_i) = \beta_i^2\sigma_m^2 + s_i^2\sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$. Under clipping, Eq. (10)

sets β_i^{eff} to give the same variance. On the R^2-preserving branch, the composed variance is $\beta_i^2\sigma_m^2/R_{i,\text{real}}^2$, the value implied by the calibrated $(\beta_i, R_{i,\text{real}}^2)$ rather than by the generator marginal. The floor guarantees only that $\text{Var}(\varepsilon_i) \geq f\,\sigma_{\text{gen},i}^2$; it does not guarantee that adding the market term preserves the generator's tail distribution. In Supplementary Table S6, the KS pass rate fell from 87.6% in the lowest-beta quartile to 65.9% in the highest-beta quartile.

S9.1. Role of the centered full-return draw as the residual

The hybrid composition places $\varepsilon_i = s_i\tilde{g}_i^{\text{c}}$, a centered and rescaled draw of asset i's full return, in the residual slot of Eq. (4). On its face this is a category mismatch: the residual of a single-index decomposition is the idiosyncratic component of the return, orthogonal to the market, while $\tilde{g}_i^{\text{c}}$ is distributed like the centered total return, market-driven component included. This appendix states what the substitution preserves exactly, what it preserves approximately, and how the approximation degrades as the market loading grows.

The residual slot must supply three things: a zero mean, so that the intercept and market term alone set the conditional mean; the variance $\sigma_{\text{gen},i}^2 - \beta_i^2\sigma_m^2$ not already contributed by the market term; and the asset-specific distributional shape, the heavy tails and volatility clustering that motivated the single-asset generator. Centering meets the first requirement exactly (Eq. (6)), the scale of Eq. (8) meets the second exactly, and the large-T limits $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$ hold because the draw is generated independently of the simulated market path (Sec. S1), so the recovered factor structure is exact as well. Any zero-mean, correctly scaled, market-independent draw satisfies the first two requirements; the full-return draw is chosen for the third, because it carries the per-asset marginal that the single-asset validation certified. Only the shape requirement is met approximately, and two exact statements bracket the approximation.

At $\rho_i = 0$ the substitution is not an approximation: Eq. (8) gives $s_i = 1$, the market term contributes no variance, and the composed series is the generator draw itself, so the composed distribution equals the validated single-asset marginal. For $\rho_i > 0$ the composed return is the sum of two independent centered draws, and the survival of the generator's shape follows in closed form. In population, conditional on the fitted parameters, write the deviation of the composed return from its mean as $X_m + X_i$, where $X_m = \beta_i\,g_m^{\text{c}}$ is the market contribution, g_m^{c} is the market path centered at its mean, and $X_i = s_i\,\tilde{g}_i^{\text{c}}$ is the scaled asset draw; their variances are $v_m = \rho_i\,\sigma_{\text{gen},i}^2$ and $v_i = (1-\rho_i)\,\sigma_{\text{gen},i}^2$. Independence factors every cross term in the fourth moment:

$$\begin{aligned}
\mathbb{E}\big[(X_m + X_i)^4\big] \;&=\; \mathbb{E}(X_m^4) + 4\,\mathbb{E}(X_m^3)\,\mathbb{E}(X_i) \\
&\quad + 6\,\mathbb{E}(X_m^2)\,\mathbb{E}(X_i^2) + 4\,\mathbb{E}(X_m)\,\mathbb{E}(X_i^3) \\
&\quad + \mathbb{E}(X_i^4) && \text{(independence)} \\
&=\; \mathbb{E}(X_m^4) + 6\,v_m\,v_i + \mathbb{E}(X_i^4). && (\mathbb{E}X_m = \mathbb{E}X_i = 0)
\end{aligned} \tag{S43}$$

Writing κ_m and κ_{gen} for the excess kurtosis of the market path and of the generator draw, the component fourth moments are $\mathbb{E}(X_m^4) = (\kappa_m + 3)\,v_m^2$ and

$\mathbb{E}(X_i^4) = (\kappa_{\text{gen}} + 3)\, v_i^2$, where rescaling by s_i leaves excess kurtosis unchanged. The excess kurtosis of the composed return follows by dividing Eq. (S43) by the squared total variance:

$$
\begin{aligned}
\kappa(X_m + X_i) \;&=\; \frac{(\kappa_m + 3)\, v_m^2 + 6\, v_m\, v_i + (\kappa_{\text{gen}} + 3)\, v_i^2}{(v_m + v_i)^2} \;-\; 3 \\
&=\; \frac{\kappa_m\, v_m^2 + \kappa_{\text{gen}}\, v_i^2 + 3\,(v_m + v_i)^2}{(v_m + v_i)^2} \;-\; 3 && \text{(collect terms)} \\
&=\; \kappa_m\, \rho_i^2 \;+\; \kappa_{\text{gen}}\, (1 - \rho_i)^2. && (v_m + v_i = \sigma^2_{\text{gen},i})
\end{aligned}
\tag{S44}
$$

The generator's excess kurtosis survives composition with weight $(1 - \rho_i)^2$ and the market path's enters with weight ρ_i^2; a simulation check with independent generator draws matched Eq. (S44) to within 0.03 in mean excess kurtosis at $\rho_i \in \{0.1, 0.3, 0.5\}$. For a low-loading asset the composed tails are essentially the generator's: at $\rho_i = 0.10$ the generator's kurtosis retains weight 0.81. As ρ_i grows the composed shape moves away from the asset's own marginal, which is the pattern in the documented diagnostics: the hybrid tracks the generator's kurtosis across the β range while the Gaussian SIM does not (Supplementary Fig. S2), and the hybrid KS pass rate declines from the lowest to the highest β quartile (Supplementary Table S6).

The residual-fit methods provide the exact-orthogonal alternative: they fit a generator to each asset's OLS residual series, contain no market component to double count, and are compared with the hybrid in Table 3; fitting JumpHMM to residuals requires converting each residual series to a synthetic price path by cumulative compounding and a second fit per asset. In summary, $\tilde{g}_i^{\text{c}}$ is not a residual in the orthogonal-decomposition sense; it is a carrier of the validated marginal shape. The substitution fixes the mean, the variance, and the recovered factor loadings exactly, reproduces the asset's marginal exactly in the $\rho_i \to 0$ limit, and dilutes the generator's shape with weight $(1 - \rho_i)^2$ as the market share grows; the clipped and R^2-preserving branches of Algorithm 2 handle the region where the dilution is largest.

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